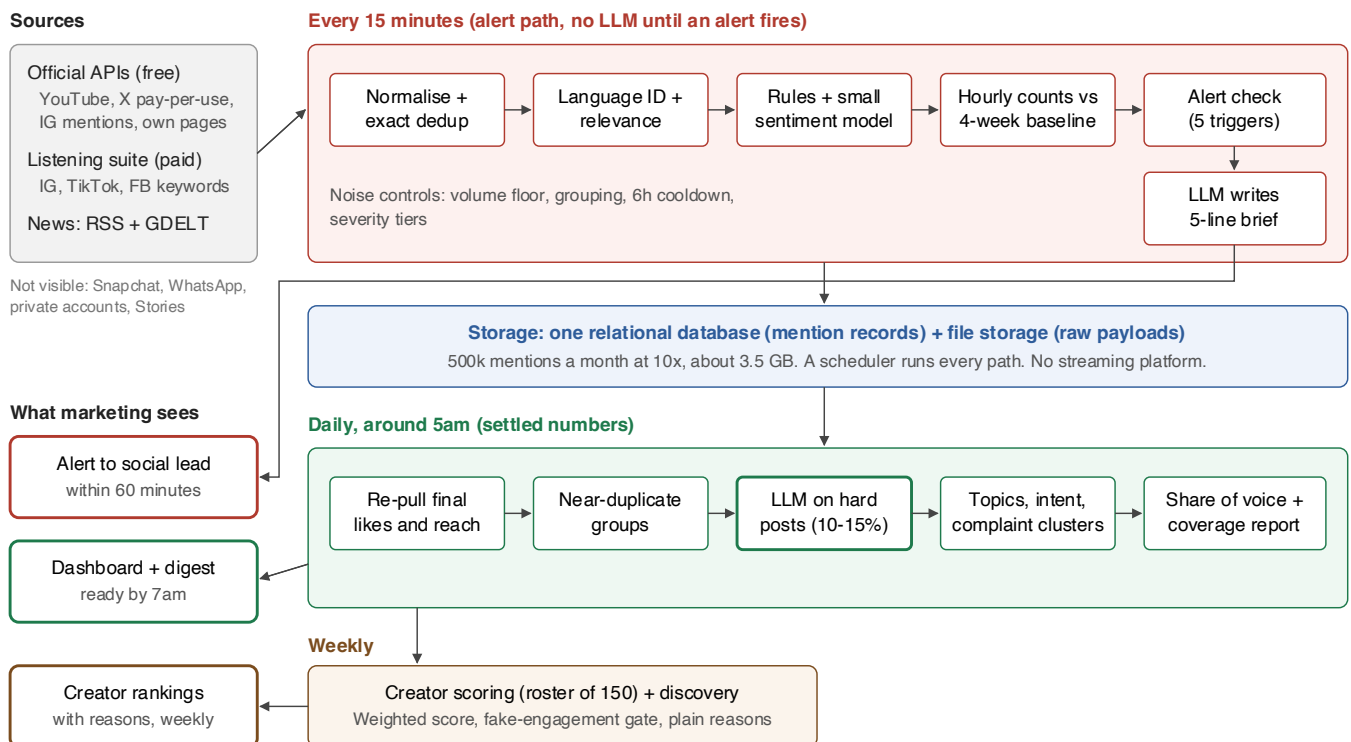


Social Listening for a GCC Fashion Brand: System Design

Today, competitor and sentiment checks are done by hand, one platform at a time. The system replaces that with a dashboard every morning and an alert to the social lead within an hour of a problem starting, across English and Arabic posts on 5 platforms and news.

Five Point Summary

1. Alerts every 15 minutes, everything else daily, creators weekly.
2. Official APIs where they exist + one vendor for the rest, no scraping (ToS risk).
3. LLM handles only the 10 to 15% (targeted, observed 70-80% in Part B without optimization) of posts that the small model can't.
4. Accuracy is measured from day one against a 600-post bilingual gold set.
5. One database and a scheduler, and data access is over 90% of the cost.



Within 60 minutes	By 7am	About \$0.005 to \$0.02
from a problem starting to an alert reaching the social lead*	the dashboard and digest show yesterday's final numbers	About \$2.5k to \$10k a month (over 90% of it data access, under \$200 (if 10-25% target achieved) for AI) ÷ 500k mentions = \$0.005 to \$0.02 per mention.

*depending on vendor freshness: has to be conformed from the vendor

Part B builds the sentiment and topic slice: English, Arabic and mixed posts run through a small model, with Gemini on the hard ones, and are scored against 300 labelled posts, with a note on where it fails.

CODE AND README: <https://github.com/karankhatavkar/mixed-language-sentiment-and-topic-cascade>

Problem Framing

What client asked for vs what they need:

Asked: a system that covers 5 capabilities across all public media and produces dashboards, alerts and recommendations.

Need:

- Each morning, tell marketing what changed in how people talk about us and the 5 rivals.

- Within the hour, warn them when something starts going wrong.

What to build (Order by Priority):

1. **Monitoring of social platforms:** everything else depends on it.
2. **Sentiment and share of voice:** drives both the morning view and the alerts.
3. **Trends and pain points:** catches complaints before they spread.
4. **Creator evaluation:** runs weekly, because creator numbers don't change daily.
5. **Creator discovery:** reuses the mention stream and the same scoring.

Not worth building:

- Our own scrapers - break platform terms, carry legal risk
- Everything in real time - only alerts need it. Likes and reach can be analysed overnight.

What "same day" means:

- Alerts reach the social lead within 60 minutes of a problem starting.
- The dashboard is ready by 7am local time with yesterday's final numbers.
- **Every 15 minutes:** pull new posts, score them with a small model, and check for alerts.
- **Daily at 5am:** settled numbers, the LLM analysis on hard posts, topics and share of voice.
- **Weekly:** creator scores and recommendations.

Questions and assumptions:

1. **Which platforms matter most?** Instagram, TikTok, X, YouTube and news.
2. **What is the monthly budget?** Low thousands of dollars.
3. **Is a paid data vendor OK?** Yes, alongside official APIs.
4. **What should alerts cover?** Volume spike, sentiment drop, high-reach negative post, new complaint cluster and crisis keywords.
5. **Do creators share their audience stats?** No, so the design uses public data only.

Data acquisition and coverage

Platform	Source	Cost	Get	Can't get
YouTube	Official API	Free (quota)	Videos, comments, stats	Others' audience data
Instagram	API + suite*	Free + vendor	Mentions, Hash tags	Keyword search, Stories
TikTok	Suite* + login	Vendor + free	videos, roster creator stats	Full comments
X	API	\$1,500/mo (unverified)	Last 7 days of posts	Older archive
Facebook	API + suite*	Free + vendor	competitor pages	Personal profiles, groups
News	RSS + GDELT	Free	Arabic and English news	Paywalled full text

*Suite: one listening vendor (e.g. Brandwatch, Talkwalker, Meltwater), about \$500 to \$10k+ a month.

Ruled out:

- **Snapchat.** No public API for listening.
- **WhatsApp.** It's private, encrypted messaging.
- **Scraping any platform,** for the ToS and legal reasons

Data Paths:

Every 15 min: pull → normalise → exact dedup → language ID → relevance → fast sentiment → hourly counts → alert check → LLM brief

Daily 5am: re-pull metrics → near-dup groups → LLM on hard posts → topics and intent → share of voice → dashboard tables

Record **Sample:** <https://github.com/karankhatavkar/mixed-language-sentiment-and-topic-cascade/blob/main/mention-record.json>

Volume and storage:

- 50k mentions a month today, 500k at 10x: about 17k a day, 700 an hour, up to 7k an hour in a spike.
- About 7 KB per mention with the raw payload: 3.5 GB a month, 40 GB a year.
- **Decision:** relational database for mention records, file storage for raw payloads, and a scheduler.
- Raw payloads are kept, so any post can be re-processed when a model changes.
- Traded away: past roughly 100x this would need revisiting.

Duplicates:

- Exact (every 15 min): the same platform and post ID updates the record and never creates a new one.
- Near-duplicate (daily): news and posts that are over 90% similar : The earliest post is the original.
- Traded away: some separate posts with near-identical wording get merged.

Models and evaluation

Part B builds the sentiment and topic slice of this design. Models marked (Part B) are the ones it runs.

- **Sentiment:** Cardiff XLM-RoBERTa Twitter sentiment, one model for English and Arabic (Part B).
- **Topics:** mDeBERTa zero-shot over a fixed topic list (Part B). Later, marketers can edit the list.
- **Intent:** Jev from TypeSafe AI, a fast classifier with calibrated confidence, over a fixed list: complaint, question, praise, purchase intent. Arabic support to be checked against the gold set.
- **LLM:** Gemini Flash-Lite for hard posts, Gemini Pro as the judge for topic labels (Part B).

Label set: Positive, negative, neutral, mixed. Mixed is for posts that are genuinely both, like "love the dress, delivery was a nightmare". Sarcasm is a separate yes/no flag, because a sarcastic post is usually just negative.

Measuring accuracy with no labels

- Gold set: 600 posts, about 200 each of English, Arabic and mixed, across all platforms, with about 20% hard cases (sarcasm, emoji-only, competitor comparisons).
- Two bilingual people from the social team label them independently, and a third settles disagreements. Their agreement (Cohen's kappa) is the ceiling.
- 200 posts tune thresholds, 400 are held back for reported numbers.
- After launch: 100 fresh posts labelled a week, plus "wrong label" clicks on the dashboard.

Part B uses the same method at a smaller size: 300 posts

Metrics

Reported separately for English, Arabic and mixed:

- **Macro-F1:** overall quality, not inflated by the many neutral posts.
- **Negative precision and recall:** alerts depend on catching negatives without false alarms.
- **Confusion table:** which labels get mixed up.
- **Topic F1:** per topic and macro.
- **Calibration:** does 0.9 confidence mean right 9 times out of 10? LLM routing depends on it.
- **LLM call rate and cost per 1,000 posts.**

I used these in Part B, except calibration, to compare three setups: small model only, LLM only, and the cascade. Results:

<https://github.com/karankhatavkar/mixed-language-sentiment-and-topic-cascade/tree/main/results>

Share of voice

Cover: keywords, brand-name ambiguity, relevance classifier, coverage imbalance, sampling from vendor.

SoV (simplified) = brand's mentions ÷ mentions of all 6 brands × 100, over a rolling 7 days, refreshed daily.

- Counts only relevant, unique, non-bot mentions from sources covered all week.
- Decision: count-based, with the sentiment split beside it ("30% overall, 45% of negative"). Reach-weighted swings 20 points on one viral post.

Traded away: big and small accounts count the same. Reach is shown separately.

Alerting

Triggers: volume spike, sentiment drop, high-reach negative post, new complaint cluster, crisis keywords (boycott, scam, unsafe, and Arabic equivalents).

Noise controls:

- Baseline is the same hour of the week over the last 4 weeks, and nothing fires below 30 mentions.
- One incident sends one alert. The same issue waits 6 hours unless it gets worse.
- Critical goes out instantly, medium to the alert channel, low to the morning digest.
- Spikes during our own campaigns are marked "likely campaign".

Example brief (LLM, 5 lines max):

🔴 Negative spike on delivery: 210 mentions in the last hour vs a usual 40, 68% negative. Why: mostly "order not arrived" after Friday's sale. Top post from a 48k-follower account. Where: Instagram and X, mostly UAE. TikTok partially checked. Suggested action: post a delivery update. [Top 3 posts] 👍👍

Usefulness: at least 70% 👍, alerts per week, incidents the team found that we missed, and time to acknowledge. Thresholds start rough and are tuned every 2 weeks.

Creator scoring and discovery

Features (each 0 to 100, as a percentile within the roster):

- Brand fit (30%): share of posts on fashion and lifestyle, how often they post for our 5 competitors.
- Audience relevance (30%): share of comments from GCC accounts, size tier that suits the campaign.
- Performance (25%): engagement rate vs creators of the same size, past collaborations with us vs their own average.
- Sentiment (15%): sentiment of comments on their posts and of what they say about us.

Decision: Score = 0.30 × Brand fit + 0.30 × Audience + 0.25 × Performance + 0.15 × Sentiment.

Fake or bought engagement: A creator is flagged if several of these show up (3 of 5 signals fire):

- sudden follower jumps with no viral post to explain them (follower jump more than 10% of current)
- engagement rate far above creators of the same size (above 3× the size-tier median)
- generic or copy-paste comments ("nice 🔥", repeated emoji) (above 30%)
- commenters with no posts, no profile picture or brand-new accounts
- audience suddenly shifting to countries outside the GCC

A flagged creator's score is capped and marked "check before booking".

Discovery: each week, authors in the mention stream who aren't on the roster, with several relevant posts and a GCC audience, go through the same score. The output is a short "suggested creators" list. Creators who often post about competitors are worth a look first.

Observability and cost

Tracing: every mention gets a trace_id at capture. Each stage writes its timestamp, model version and result against it.

Example: captured 08:52 from Instagram → mixed EN/AR, relevant (0.94) → negative (0.91, small model, no LLM) → counted in alert al_003 at 09:00 → topic "delivery" confirmed in the 5am batch → shown on the dashboard.

Cost per month = data access + compute + (mentions × LLM share × cost per LLM call)

At 10x (500k mentions a month):

- LLM: about \$0.25 per 1,000 posts sent. At 10 to 15% of posts that is about \$15 to \$20.
- Compute for the small models: about \$50 to \$150.
- Data access: X about \$1,500, plus the suite.
- All in: about \$2.5k to \$10k, or \$0.005 to \$0.02 per mention. Data access is over 90% of it, vendor choice matters more than the model choice.

Budget guard:

- Daily LLM spend cap, with a warning at 80%.
- When the cap is hit, low-priority posts wait for the next daily batch. Alert briefs always go out.
- Monthly X and vendor reads tracked against plan limits.

What we watch:

- LLM call rate (a jump means the small model is struggling) and daily cost vs budget.
- Source freshness: time since the last successful pull, and expected vs received volume.
- Pipeline lag from posted_at to processed_at. The alert promise needs it under 15 minutes.
- Label drift: weekly agreement with the gold set, and how often the small model and the LLM disagree.

Part B and limits

Built: the sentiment and topic slice. Two free models (XLM-R for sentiment, mDeBERTa zero-shot for 8 fashion topics) run on every post, and Gemini Flash-Lite takes the hard ones. Tested on 300 public posts (English, Arabic, mixed) and 234 YouTube comments from 3 SHEIN haul videos, labelled by hand. Repo: <https://github.com/karankhatavkar/mixed-language-sentiment-and-topic-cascade>

YouTube results (overall F1 / negative recall / LLM calls / cost per 1k posts):

- Small models only: 0.44 / 0.62 / 0% / \$0
- LLM only: 0.91 / 0.98 / 100% / \$0.25
- Cascade: 0.81 / 0.95 / 76% / \$0.19

Where it fails:

- The small model misses Arabic complaints: 27 of 61 Arabic negatives on YouTube read as positive.
- The cascade sends 70 to 76% of posts to the LLM, far above the 10 to 15% the design targets.
- YouTube labels started as Flash-Lite suggestions that I reviewed, so the LLM-only score is likely flattered.
- Public topic labels come from a Gemini Pro judge.

With four more weeks: label 600 of our own posts as the real gold set, tune the routing threshold and check calibration, add Arabizi, and build the alerting layer on this pipeline.

Left out on purpose: creator scoring, alerting, the dashboard and live collection. Time spent: 8 hours.

Ran out of time for: Optimization of cascade and small models used.